

Autonomous AI Bot for Self-Healing and Proactive Server Health Management in Distributed Systems

Comments for Future Enhancements

- The presentation was clear and well-structured, showing a good understanding of the topic.
 - The evaluation criteria should be improved by defining clear baselines or comparison models.
 - Clarify how real-time performance will be maintained once ML and RL components are integrated.
 - Provide a stronger justification for the synthetic data generation process to ensure fault scenarios are realistic and reproducible.
 - Discuss potential security and ethical implications of allowing AI to perform autonomous recovery actions.
 - Explain the expected technical limitations or challenges during implementation.
 - Justify why Reinforcement Learning (RL) was selected for the healing module instead of other approaches.
 - Clearly state the main research gap being addressed and why it is important.
 - Ensure that the synthetic data used in experiments closely represents real-world failure scenarios.
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- The project needs stronger **real-world validation** — currently, it relies mainly on datasets and synthetic data. A small-scale deployment or simulation (e.g., using Docker or Kubernetes) would help verify how the AI bot performs in unpredictable, live environments.
 - The **reinforcement learning module** should include clear safety controls or fallback mechanisms to prevent harmful or unnecessary recovery actions during training or testing.
 - The **explainability layer** (using LIME, SHAP, and LLMs) should be tested with real users or admins to confirm that the explanations are understandable and actually improve decision trust.
 - More clarity is needed on **how performance will scale** as the number of nodes increases — showing test results or benchmarking data would make the findings stronger.
 - The “adversarial-aware” claim should be supported with an example or brief demonstration of how the system detects or defends against model attacks or false

inputs.

- Since the system is autonomous, it's important to discuss **ethical and operational safety measures**, such as human oversight or rollback procedures in case of incorrect actions.
- The **research gap** should be stated more precisely — for example, emphasizing the lack of an integrated framework that combines anomaly detection, self-healing, and explainability in distributed systems.

Also,

- Include a clear **system architecture diagram** in the report or presentation to visually explain how the ML, RL, and XAI components interact within the self-healing framework.
- Define specific **evaluation metrics** for both the anomaly detection accuracy and the healing success rate (e.g., downtime reduction %, false positive rate, MTTR). This would make the results easier to quantify and compare.
- Consider adding a **timeline or roadmap** for implementation stages (data preparation, model training, testing, integration) — this helps demonstrate planning and research management skills.
- Briefly discuss the **hardware or infrastructure requirements** for deployment — it will help show the project's practicality and scalability.
- Finally, in your conclusion, summarize the **expected real-world benefits** (e.g., cost reduction, improved uptime) to clearly connect the research outcomes to industry relevance.